

# FOFC in 2025

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## 1 Introduction

During the 21st Century, the term *Final-over-Final Condition (FOFC)* has been used to describe a type of rare or unattested word order patterns. The earliest and most basic formulation of FOFC is as follows (Holmberg, 2000, pg. 124) :

- (1) *The Final-over-Final Condition (FOFC) (unrestricted version):*  
If a phrase  $\alpha$  is head-initial, then the phrase  $\beta$  immediately dominating it is head-initial. If  $\alpha$  is head-final,  $\beta$  can be head-final or head-initial.

This formulation is informal and does not place any restrictions on its nature or application. As can be seen in (1) and subsequent definitions, FOFC is defined "relative to the string *under a given structural description*" (Roberts, 2017b, pg. 31). The morphism between surface string and underlying structure is crucial to analysing FOFC. For the two heads  $\beta$  and  $\alpha$  and the complement head of  $\alpha$ ,  $\gamma$ , the surface strings of note have the linear order  $\alpha - \gamma - \beta$ , also referred to as 2-3-1, with the numbers representing the relative structural height of each head. The linear order  $\alpha - \gamma - \beta$  alone does not create a FOFC violation, but this must be assessed under a given structure. Excluding movement, the basic structural possibilities for (1) are shown below:

- (2) a.  $\beta P$   
 $\swarrow \searrow$   
 $\beta \quad \alpha P$   
 $\swarrow \searrow$   
 $\alpha \quad \gamma$   
Initial-over-initial
- b.  $\beta P$   
 $\swarrow \searrow$   
 $\alpha P \quad \beta$   
 $\swarrow \searrow$   
 $\gamma \quad \alpha$   
Final-over-final
- c.  $\beta P$   
 $\swarrow \searrow$   
 $\beta \quad \alpha P$   
 $\swarrow \searrow$   
 $\gamma \quad \alpha$   
Initial-over-final
- d. \*  $\beta P$   
 $\swarrow \searrow$   
 $\alpha P \quad \beta$   
 $\swarrow \searrow$   
 $\alpha \quad \gamma$   
Final-over-initial

The name Final-over-final Condition arises from the implicature that a lower head-final phrase implies the immediately dominating phrase is also head-final.

Definition (1) cannot be taken as an absolute universal under any theoretical

approach. This can be straightforwardly demonstrated for standard Minimalist analyses of German verb-phrases.

- (3)  $[_{VP} [_{PP} \textit{in die Stadt} ] \textit{gehen} ]$   
           in the-F city      go-INF  
       ‘to go into the city’

In (3), a head-initial Preposition Phrase is immediately dominated by a head-final Verb Phrase. One possible approach for data such as (3) is to reanalyse the structure that has been assigned. In cases where this structure is appropriate, this suggests modifying definition(1) to a more restrictive version, either in terms of the scope of FOFC or the nature of the heads it applies to.

This leads to a situation where understandings of FOFC can diverge significantly. FOFC requires pretheoretical knowledge relating surface strings to underlying structures. Then, different restrictions can be made to its definition. This means that different data can be considered a FOFC violation under some analyses, but not for others. This work aims to review the different approaches that have been taken to FOFC in the past 25 years, and see how far these can be reconciled.

The structure of the paper is as follows: Section 2 looks at the a wide range of analyses of FOFC and the viewpoints that are relevant to these, noting the similarities between many approaches. These approaches have been grouped as representing FOFC as either a universal on structure building, a processing universal, a statistical tendency or a universal constraint that can be overwritten. This grouping captures some information about the papers into a typology of approaches, but is primarily arranged for organisation in reading rather than deep conceptual differences. The section will finish with an evaluation of the compatibility of these approaches, and sketching a vague outline of what there is general agreement on. Section 3 looks at literature which does not propose a new approach to FOFC, but looks at empirical data in a given language or domain. It covers the range of languages studied in relation to FOFC, as well as observations about information structure, other disharmonic orders, morphology, acquisition and multilingualism, and modality. There will then be another evaluation of how these aspects broaden the general picture of the causes of FOFC presented.

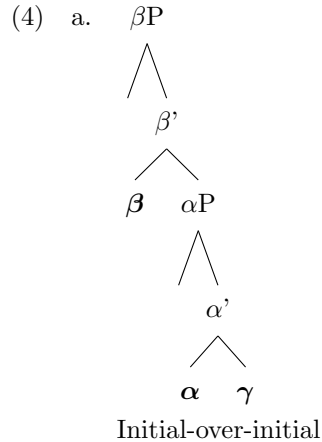
## 2 Typology of approaches to FOFC

### 2.1 Range of approaches

This section reviews a number of theoretical approaches to FOFC, abstracting away from syntactic frameworks to focus on the proposed factors that are directly responsible for FOFC. The two major loci for the factors are asymmetry and the nature of and relations between categories.

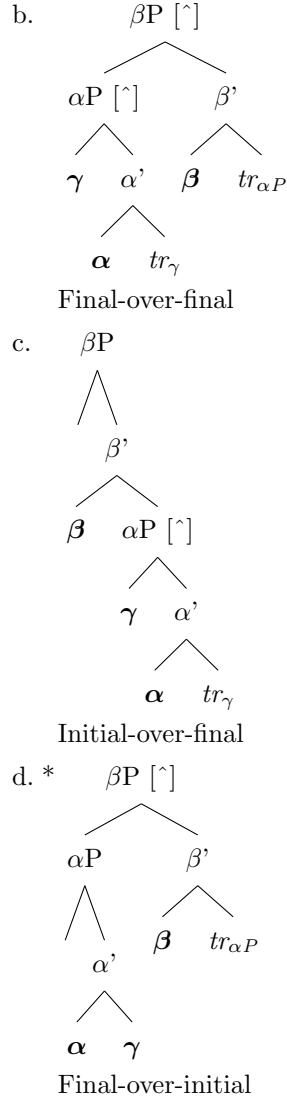
FOFC is useful for investigating (a)symmetry in syntax. Work under the Minimalist Program (Chomsky, 1995) views narrow syntax as symmetric, whereas work under the Linear Correspondence Axiom (Kayne, 1994) views it as fundamentally asymmetric. Most of the earliest studied asymmetries hold between heads and their specifiers or complements, which is in contrast to FOFC which should hold between heads. The asymmetry in (1) is between head-initial and head-final ordering, but relates not to these heads but rather the linearisation possibilities for immediately dominating heads. The approaches covered in this section attempt to present different views on what asymmetries cause FOFC, and why they should arise in a non-stipulative manner.

Example (3) shows that pursuing an unrestricted FOFC does not seem insightful, and we need to somehow limit its applications. The approaches investigate different properties of categories of syntactic items in order to explain this. Highly influential early work on FOFC by Biberauer et al. (2014) introduces the notion of Extended Projection (EP) (Grimshaw, 2000)<sup>1</sup>. This proposes that lexical categories share a categorial projection with their related functional heads, which can be represented with the features  $[\pm V]$  and  $[\pm N]$ . By hypothesis, FOFC applies only within EP, so in (3) the nominal EP is head initial, and this is dominated by a head-final verb, but there is no violation as the verb does not belong to the same EP. Within EPs, the lowest head can be linearised as head-initial or -final. Higher heads can freely be head-initial but may only be head-final if the head immediately below is also head-final. This creates an asymmetry between the lowest heads and higher heads. The allowed linearisation patterns are also monotonic, as head-finality within a given EP must be contiguous and start at the bottom. In the LCA, head-final order is derived from *roll-up* movement of a complement to the specifier position. The derivations of (2) are given below, where  $[\hat{\cdot}]$  is a feature representing head-finality:




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<sup>1</sup>although there are differences between the definition used by Grimshaw and the one used in FOFC literature.



In this way, FOFC may be able to illuminate interesting properties of how humans form and use categories in language, which have been further investigated in a range of approaches.

### 2.1.1 Universal on structure building

A number of approaches have continued the path of Biberauer et al. (2014). Hein and Murphy (2022) use the notion of mixed extended projections (Borsley and Kornfilt, 1999) and evidence from West African languages to suggest FOFC holds in cases of nominalisation. This requires a rethink of the separation of the verbal and nominal domain central to the above approach. Cinque (2023)

considers monotonicity important, but views FOFC as more likely a strong statistical tendency rather than a universal over the domain it applies. In this way, monotonicity can be seen as a general factor in human cognition rather than intrinsically related to the properties of linguistic categories.

Biberauer (2017, et subq.) investigates the explanatory power of the 2010 approach in detail. It is noted that while the LCA helps with the exposition of FOFC, it is possible to maintain the same outputs without reference to roll-up movement or the theoretical assumptions required in the LCA. Monotonicity does not arise from a single feature  $[\wedge]$ , but can be seen as relevant in human cognition in many domains within and outside of language, as will be discussed in the next section. This approach uses the notion of Input Generalisation (Roberts, 2017b) to form a *None>All>Some* learning path. This means that first, no heads have a given property, in this case  $[\wedge]$ . Then when this property is acquired, it is first maximally generalised to heads within a category. If this overgenerates, the domain of the property can be reduced to a subset of heads. This captures the empirical fact that harmonic head-final languages seem relatively unmarked and can be very stable.

The account looks at elements which are not predicted to be subject to FOFC. There are many elements known as Sentence-Final-Particles (SFPs) which are in final position in the clause, but appear to take scope over the verb. These exist in a very wide typological range of languages, and have varying functions, but the best investigated examples come from Chinese varieties like in the example below.

- (5) *Tā chī-le fan le*  
 3SG eat-PERF food PERF  
 ‘He has eaten’  
 (Paul, 2014, pg. 86)

The account states that only elements within an EP should be subject to FOFC, predicting that *le* must be categorially distinct from *chī*, and not possessing the feature  $[+V]$ . In this way, SFPs are not predicted to form a homogeneous class, and FOFC-compliance should not be seen as the default. The formal reasons an element may not belong to the extended projection include that the element has no formal features, categorially distinct formal features, or is just a PF-reflex not in the numeration. The properties of such elements often include that they do not display  $\phi$ -agreement, they are semantically underspecified, they are less grammaticalised, they can appear in multiple positions<sup>2</sup>, and they have differing diachronic paths to the elements in the lexical extended projections. These properties are not found in the FOFC-obeying elements such as Germanic

<sup>2</sup>Although these positions are all within the verbal domain, so there is contention that these should reasonably be considered categorially distinct (Erlewine, 2017).

auxiliaries.

DelBusso (2020) investigates within the framework of Optimality Theory (OT), particularly using Stringency Theory, which looks at relations between constraints. Within an EP, there is one constraint violated for any head-initial phrase, and  $n$  constraints  $\text{Head-Left}_0 \dots \text{Head-Left}_{n-1}$  corresponding to  $n$  heads  $H_0 \dots H_{n-1}$  in an EP, where  $H_1$  is the lowest head.  $\text{Head-Left}_0$  is violated by any head-final phrase, and  $\text{Head-Left}_1$  is violated by any head-final  $H_1 \dots H_{n-1}$ . This continues until the final constraint, where  $\text{Head-Left}_{n-1}$  is only violated by a head-final  $H_{n-1}$ . The head-final constraints are in a Stringency relation with one another, meaning a subset of the candidates that survive one survive another. In this case, the constraint relating to the lowest head in the EP is the most stringent, and constraints relating to higher heads are successively less stringent. The only linear orderings that can be optimal under a given ranking of the constraints are ones which do not violate FOFC. This has a *None > Some > All* learning path, different to before. This captures the diachronic possibilities for language change under FOFC.

Erlewine (2017) takes the domain of FOFC not to be extended projections, but rather phases. Using evidence from the scope of Chinese SFPs, it is proposed that SFPs must occur in the position of phase heads, and their complement is a spell-out domain. Phase heads can be initial or final, but there cannot be a switch from initial to final within a phase. To account for cases where FOFC applies between a lexical verb and higher verbs, typically proposed as a phase boundary, there is an appeal to head-movement which expands the spellout domain. This raises interesting questions as to the position and nature of phases, as well as their purpose in cognition.

Some approaches restrict the domain of FOFC to linearisation at the Phonetic Form (PF). This suggests significant differences compared with LCA based approach, where the derivation of structure with a total order takes place incrementally, although there may be different implications for models which have a symmetrical narrow syntax. Duguine (2022) investigates Basque, which has a verbal EP which is head-final at the bottom but head-initial in certain higher projections dependent on Information Structure. It is shown that surface structures with preverbal clausal embedding are ruled out with an overt high initial head, but are permissible when the presence of such a head is assumed but is phonetically null<sup>3</sup>, suggesting FOFC-violating structure may be built as long as it is modified before linearisation takes place, which has implications for the strategies to avoid this.

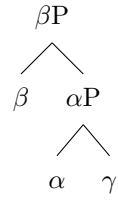
Sheehan (2013) suggests a modification to Kayne's LCA, noting that the asymmetries between a specifier and a head are empirically motivated, but be-

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<sup>3</sup>One case which has been used as evidence against FOFC purely at PF is found in Hindi, in which initial complementisers can be realised as phonetically null, but doing so does not render the structure acceptable in FOFC-violating positions (Sheehan et al., 2017).

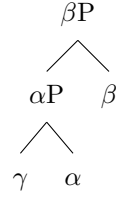
tween a head and a complement less so. It is proposed that the ordering of two heads in a c-selection relation is parametrised. If transitive relations between a series of heads does not give a total ordering, asymmetric c-command applies as a last resort. The derivations of (2) are given below, where  $\beta$  asymmetrically c-commands  $\gamma$ , and c-selection relations hold between  $\beta$  and  $\alpha$ , and  $\alpha$  and  $\gamma$ , indicated here with  $<$  or  $>$ .

- (6) a.  $\beta < \alpha, \alpha < \gamma$



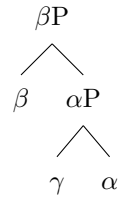
Initial-over-initial via c-selection

- b.  $\beta > \alpha, \alpha > \gamma$



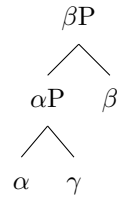
Final-over-final via c-selection

- c.  $\beta < \alpha, \alpha > \gamma$



Initial-over-final -  $\beta$  and  $\gamma$  are not ordered transitively by c-selection  
so  $\beta < \gamma$  via asymmetric c-command

- d.  $\beta > \alpha, \alpha < \gamma$



Final-over-initial -  $\beta$  and  $\gamma$  are not ordered transitively by c-selection  
so  $\beta < \gamma$  via asymmetric c-command, resulting in the order  $\alpha - \beta - \gamma$ .



In this way, FOFC violating orders cannot result from linearisation.

Richards (2016) situates FOFC within a wider theory attempting to unify patterns of linearisation and prominence in prosody. Heads which select another phrase enter into *Selectional Contiguity* with its head, meaning that linear adjacency must be present at some point in the derivation. This can create consistent head-initial orderings using the LCA, as there will be linear adjacency between categories. It is proposed that conditions of prosodic prominence in languages are parametrised, and this might favour head-final linearisation. The LCA gives an ordering of two heads in an asymmetric c-command relationship, but it is proposed that languages might make available the option of *untethering* this linear order, allowing for different potential orderings, if this is motivated by contributing to selectional contiguity. As the bottom pair of elements in an EP start with no asymmetric c-command relations, they are untethered by default and can be head final if the prosodic conditions allow for this. The asymmetry between Final-over-Initial and Initial-over-Final orders arises again because head-finality can only start at the bottom of an EP, and a head-initial category cannot dominate a head-final one as there is no motivation for the higher category to untether as this does not help with Selectional Contiguity. In the reverse case, Selectional Contiguity can later be broken in cases of head movement, causing an Initial-over-Final structure.

For independent reasons, Richard’s theory requires SFPs to belong to the class of complementisers. There is an asymmetry between subordinating and interrogative C heads, in that subordinating ones will never untether as this does not help Selectional Contiguity, whereas interrogative ones may, if the prosodic conditions allow this, to form Selectional Contiguity with wh- elements. It is predicted that only phase heads will be able to violate FOFC this way, as the adjacency to the lower head is rescued by being in a separate spellout domain.

These approaches see FOFC as providing an insight into human categorisation and computation. There are different mechanisms proposed for this, but these may be more reconcilable than it appears. Concepts such as Extended Projection, Phase, and Spellout Domain come up repeatedly, as well as the relation between narrow syntax and linearisation, but the ontology of these is strongly debated in the literature. Investigation into FOFC might be able to unify and refine the properties of these, giving us wider insight into not only FOFC but human syntactic and cognitive capabilities more generally.

### 2.1.2 Processing universal

Cecchetto (2013) builds on work suggesting that left-right asymmetries amount simply to a ban on rightward movement, with narrow syntax being symmetrical (eg. Abels and Neeleman, 2012). Cecchetto notes the similarities in terms of processing between the dependency of a higher to a lower head, and that of a filler to a gap, in that the latter category in each pair depends on the former. Both FOFC and the Right-Roof Constraint (Ross, 1967) can be seen as backward dependencies for the parser, and Cecchetto proposes these must be

phrase bound, unlike forward dependencies which can be cross-phrasal. This rules out dependencies between heads when the higher one linearly follows the lower one. For cases like the German DP, it is argued that because D never moves to V, V selects DP rather than V, so this does not lead to a cross-phrasal dependency. A similar analysis must apply for final-over-final constructions<sup>4</sup>, leading to asymmetries in selection.

Zeijlstra (2023) also takes an approach which bans rightward movement. It is proposed that if a particular structural position in a language is a movement target for at least one item, this position must always be linearised to the left of the initial position for the item, but positions which are not movement targets can be left or right-linearised. This straightforwardly rules in initial-over-initial and initial-over-final constructions, and rules out final-over-initial ones where there is the possibility for movement. Zeijlstra leaves open the question of whether final-over-final constructions can allow head movement, noting that this would always be string-vacuous and as such the debate in the literature involves more indirect arguments. This approach allows for surface FOFC violations when the higher head is never a movement target for a lower one - notably meaning there cannot be any  $\phi$ -agreement in the higher head, allowing SFPs. This approach also explains the distribution of specifiers: right-linearised ones are possible, but the majority are targets for A- or A-bar-movement meaning they must be left linearised.

Medeiros (2021) proposes a radical rethinking of syntax to a *Universal Parser as a Universal Grammar* model, using trees from computer science rather than linguistics. It is proposed that there is a universal semantic hierarchy, which corresponds to the post-order traversal of a tree. The postorder is obtained by traversing the tree from left-to-right, and bottom-up. The linear order arises from the pre-order traversal of a tree, which is left-to-right and top-down. Given three heads 1, 2, 3, corresponding to earlier *beta*,  $\alpha$ ,  $\gamma$ , given the hierarchy is fixed so that the postorder must be  $1 < 2 < 3$ , the following pre-orders are possible:

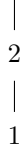
(7) a.  $1 < 2 < 3$



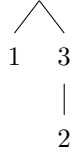

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<sup>4</sup>Cecchetto argues that the predictions of this have empirical support, notably the presence of phrasal morphology to avoid FOFC violations and the absence of fusional morphology in OV languages.

b.  $3 < 2 < 1$  3



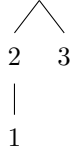
c.  $1 < 3 < 2$



d.  $3 < 1 < 2$  3



e.  $2 < 1 < 3$



Of the 6 possible permutations, only  $2 < 3 < 1$  is impossible. This corresponds to the orders that can be stack sorted into a  $1 < 2 < 3$  order, with 1 being deepest. Intuitively, if the input is  $2 < 3 < 1$ , everything must be placed into the stack so that 1 can be bottom of the output. However, 3 is then top of the stack, meaning 2 cannot be placed on the output next. This theory unifies a number of syntactic phenomena, but it does lead to results which are different from mainstream syntactic theory. Related to FOFC, a SFP must be lower on the semantic hierarchy rather than higher, meaning that it takes position 3 in a  $1 < 2 < 3$  pattern rather than a FOFC violation. Medeiros appeals to the semantics of such particles to explain this.

These approaches particularly engage with the issues of linearity as this relates to human cognition. They try to reconcile asymmetric phenomena like FOFC with symmetric views to syntax which reject models like the Linear Correspondence Axiom (Kayne, 1994). These approaches wish to portray harmonic orders as symmetrical regardless of ordering, although suggesting that backwards final-over-final dependencies amount to differences in selection introduces a significant asymmetry in itself (Sheehan, 2017). These approaches have a focus on human processing, although the key role played by movement means

that these are rooted in traditions of Generativist syntax.

### 2.1.3 FOFC as a statistical tendency

A number of approaches see FOFC as a purely statistical effect, which fits in with the significant literature on word order patterns. Many strong patterns and asymmetries in surface word order have been attested, and absolute universals for these are controversial. In these approaches, surface strings which appear to violate FOFC may have genuine underlying structures which are FOFC violations, but this is dispreferred.

Roberts (2017b) relates FOFC to the cross-categorical word order implications investigated in the tradition of Greenbergian linguistic typology (Greenberg, 1963). FOFC can be seen as a one way implication - a lower head-initial order implies a higher head-initial order, but not vice-versa. Between the two categories, head-initial orders are harmonic with one another, as are head-final orders, as this is a symmetric relation that is always permitted. Dominance is a unidirectional asymmetric relation referring to the option available with harmonic or disharmonic categories. This can be visualised in a table (Roberts, 2017b, pg. 29):

$$(8) \quad \begin{array}{|c|c|} \hline \begin{array}{c} [\beta P \ \beta \ [\alpha P \ \alpha \ \gamma P] ] \\ [\beta P \ [\alpha P \ \gamma P \ \alpha] \ \beta] \end{array} & \begin{array}{c} [\beta P \ \beta \ [\alpha P \ \gamma P \ \alpha] ] \\ *[\beta P \ [\alpha P \ \alpha \ \gamma P] \ \beta] \end{array} \\ \hline \end{array}$$

Either high head-final orders or low head-initial orders can be taken as recessive (the opposite of dominant). Appealing to the proposal that left-right word order asymmetries arise because head-finality is more marked (Kayne, 1994), Roberts concludes that head-final orders are recessive, and occur in harmonic contexts. This statement is qualified with Input Generalisation so that rigid head-final systems are not overly marked. Greenbergian implications have a focus on surface strings rather than the underlying structures, and the strongest implicatures represent overwhelming statistical tendencies rather than absolute universals.

Whitman (2008) distinguishes cross-categorical generalisations including FOFC from other constituent order generalisations, and decreases the relevance of these in synchronic factors, instead appealing to diachronic processes of reanalysis and markedness to account for an interpretation of empirical data where these are all necessarily statistical rather than universal. This approach avoids the theoretical issues associated with word-order parameters and does not risk ruling out attested structures.

Hucklebridge (2023) presents a computational model where FOFC effects arise from learnability concerns, rather than a dispreference in parameters. The

model involves a four binary parameters which initially can be set either way, as well as asymmetrical restriction banning rightward movement. The computational learner is presented with unlabelled input strings and successively updates the weightings for different parameter settings to match the presented data. In the case of FOFC violating strings, there are two parameter settings consistent with the data - 0010 and 1110. Although there are enough parameters consistent with the data, FOFC is learnt less successfully than the other word order patterns. This is because while two parameters can be set unambiguously - the final 10 - the first two can either be 00 or 11 with no strong indication of which is preferable. The learner is pulled in two separate directions, with the two parameters relying on each other. For every other word order permutation, there is a way that three parameters can be unambiguously set, and the fourth is then vacuous. Although this leads to a dispreference for FOFC violating orders, the learner is still able to acquire these, meaning that this does not constitute a banned order.

Mobbs (2015) shares Cecchetto (2013)’s observation that dependencies between heads are similar to those between fillers and gaps, noting a key difference that while a gap entails a filler, a lower functional head does not imply a higher one (giving Exceptional Case Marking constructions as an example). When the parser is presented first with a lower head, it cannot process if this is an argument of a higher predicate or not until it reaches the higher head. This is non-optimal and dispreferred unless the dependency is very local, and this also predicts FOFC to hold stronger within categories than across them.

Hawkins (2013) presents an approach to word-order patterns based upon online processing by the hearer, which places great importance on the notion of harmony. It is proposed that grammars conventionalise structures that are easier to parse, and harmonic orders minimise the dependencies between nodes in left-to-right parsing of a sentence, making them preferable as basic word orders. Under this view, FOFC does not exist, as initial-over-final constructions are not predicted as notably rarer than it, and the asymmetry is harmonic vs disharmonic<sup>5</sup>. Formulations of FOFC are seen as simultaneously too strong in disallowing surface violations and too weak in allowing unattested/strongly dispreferred initial-over-final patterns. This is supported with data from a wide typological sample accounting for areal tendencies. In the sample, *V NP T* order does not appear rarer than *TP NP V* order<sup>6</sup>, and both *C S NP* and *NP S C* orders of relative clauses with overt complementisers are unattested<sup>7</sup>, showing symmetries. This approach predicts that a larger intervening category between

<sup>5</sup>Clearly this is not the view taken by the other papers which discuss FOFC, but it does serve as an important reminder that an undue focus on surface 231 orders might obscure wider patterns of linearisation.

<sup>6</sup>The approaches in later sections which reference sentence-final particles do not see all the TPs here as the same category

<sup>7</sup>As Hawkins notes, the wording of many approaches to FOFC does not explicitly exclude the *C S NP* pattern despite its surface order, although there are approaches to relative clause word order patterns which do not make use of FOFC (eg. Cinque, 2020)

the two heads leads to an increasing dispreference for the structure, which accounts for the complete absence of large relative clause disharmonies, compared to the greater occurrence of Verb/Tense disharmonies.

These approaches can hold across relatively many theoretical frameworks, and are not challenged by the existence of cases which look to be violations on the surface. (Hucklebridge, 2023) mentions the Iterated Learning Framework (eg. Kirby et al., 2004), which proposes that weak biases are amplified over time, which might be applicable for explaining the near-absolute absence of many patterns which are claimed to be FOFC-violations despite how these are predicted possible. However, they can struggle to explain the different frequencies and behaviours of particular categories with regards to FOFC as proposed by many of the other approaches.

#### 2.1.4 Universal constraint that can be overwritten

Several approaches view FOFC as a universal constraint or a result of the interaction between different constraints, but look to situate this in a wider background of syntactic constraints, suggesting that the interaction between these can lead to FOFC being overwritten in practice. These approaches are mainly, but not exclusively, found within the framework of Optimality Theory (OT), as this investigates universal but violable constraints.

Kiparsky (2024) proposes a set of constraints grounded in semantic composition. A lambda calculus representation of the ditransitive verb *give* is given as follows:

$$(9) \quad \lambda z \lambda y \lambda x [x \text{ CAUSE } [y \text{ HAVE } z] ]$$

The lowest theta role merges with the predicate to form a new predicate, which can then recursively merge with the higher roles. As such, OV is generalised subject-predicate order. Functional heads have no theta roles and are functors instead of predicates, which precede their complements. Finally, there is also a constraint favouring harmony. This creates the following set of constraints:

- (10) a. XP<HEAD: Heads follow their complements.  
b. OP<XP: operators (functional heads) precede their complements.  
c. C<XP: subordinating operators precede their complements.  
d. HARMONY: If A dominates B, then A, B have the same headedness.

It is straightforward to see how different orders of the constraints in (10) can derive the harmonic orders, as well as a one-way implicational hierarchy for disharmonic ones such that a head-final CP implies head-final TPs, which imply

a head-final VP. The distinction between functional and lexical heads straightforwardly allows for the patterns shown across extended projections such as (3).

Philip (2013) also makes a distinction between different types of heads, in particular a class of functional heads termed *linkers*, which do not contribute their own compositional semantics, but rather mark an already existing relation syntactically and phonologically. Mandarin *de* is an example of a linker. The following constraints concerned with harmony are proposed:

- (11) a. *Head-Proximate Filter*  
 The highest head in a complete extended projection must be contiguous with the lexical head of its superordinate extended projection.
- b. *Final-Clause Constraint* A clausal dependent must follow the lexical head of its superordinate extended projection.
- c. *Head Uniformity Constraint* A functional head must match the lexical head of its extended projection in the direction of headedness.

The Head-Proximate Filter must universally take precedence over the other two. In cases where the lower head is a linker which has no semantic features, the only available base-generated orders are the harmonic ones. When the lower head involves compositional semantic information, it is possible, although dispreferred, for both disharmonic orders to arise as base generated. This gives the outlook similar to Hawkins (2013) that most FOFC formulations are simultaneously too strong and too weak.

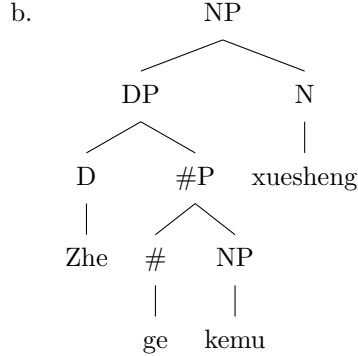
Liao (2017) looks at possible challenges for FOFC in the nominal domain in Mandarin<sup>8</sup> The most problematic cases are of the form below (Liao's (20)<sup>9</sup>):

- (12) a. *zhe ge kemu de xuesheng*  
 DEM CLF subject DE student  
 'a/the student of this subject'

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<sup>8</sup>Liao suggests that the more prominent debate about the verbal domain in Mandarin is too controversial to pose a real threat to FOFC.

<sup>9</sup>This leaves open the position of *de*



Differently from OT, Liao proposes that universal constraints like FOFC maximally hold, but can be minimally violated when this conflicts with language specific constraints, which always take priority. In this case, the proposal is that head-initial order of lexical categories only arises under case assignment, and this creates an asymmetry between the nominal and verbal domain because there is no case assigner for the complement of a nominal.

Many of these approaches consider only a limited system, where FOFC violations are never optimal. DelBusso (2020) notes this, saying that FOFC violations should be harmonically bounded. However, higher ranked constraints unrelated to FOFC could result in a FOFC-violating output being optimal in a wider system. As such, these approaches can account for superficial FOFC violations, although they struggle to delimit the range of possible violations and systematicities within these. These approaches look at the relations between FOFC and other pressures in the wider system of the grammar.

## 2.2 Evaluation of approaches

These approaches appear to vary considerably in their proposed applicability of FOFC. Much of this variation comes from differing focuses on aspects of FOFC, or theoretical differences which are wider than the likely reasons for it. In this section, I will take a look at some of the recurring ideas in the spirit of many of the papers, in order to see how reconcilable these are. I will suggest some potentially promising lines of enquiry, although I will not attempt to set out an exhaustive explanation for FOFC.

The papers have viewed the cognitive processes behind FOFC in different ways. From the perspective of a listener, the specific tasks are to delineate input strings into discrete segments, acquire the categories of these segments, and then postulate a hierarchy for the categories. When producing outputs, this hierarchy must be used and converted into linear strings. These processes complement one another. We have seen that categories with superficially similar functions across or within languages make use of idiosyncratic differences in form and



structure, making these an unsuitable candidate for content within a Universal Grammar. A more promising route of enquiry is that language acquirers use domain general capacities (third factors, under the three factor model of Chomsky (2005)) to create categories in response to the data they are presented with, in a Neo-Emergentist model (Biberauer, 2019).

Lexical items such as verbs and nouns are accessible from the earliest input the child receives, and their headedness is an early-acquired phenomena (Tsimpli, 2014)<sup>10</sup>. There can be differences in basic headedness between the basic categories, as has been seen in Chinese varieties which have a head-initial verbal domain and a head-final nominal one. Paying particular attention to the order of object and verb as one of these pairs, there is a roughly 50/50 incidence of each preferred order across typological samples of the world’s languages (Dryer, 2013). This symmetry fits in awkwardly with debates in the literature about a basic head-complement order, with different proposals favouring Head-Complement order with evidence from structure (Kayne, 1994) and in cognition (Abels and Neeleman, 2012), or Complement-Head order from structure (Haider, 2000), semantics (Kiparsky, 2024) and prosody (Tokizaki and Kuwana, 2013), both among others. However, symmetry at the bottom of lexical projections is found in FOFC.

For higher functional categories, there are many explanations for harmony in the literature, and this can be described vaguely as a preference for minimising dependencies and maintaining contiguity. The Initial-over-Final option is also not excluded by FOFC, and for this again we can appeal to some formulation of the intuitive idea that linearisation of elements in the high periphery to the left is less marked, corresponding to real-time processing. It can be noted that some examples of this order, like Germanic TOV sequences in matrix clauses, also have semantic marking in the form of topicalisation, meaning that even though this is a neutral order for information structure, it doesn’t reflect base-generation.

These factors can give a vague idea of why Final-over-Initial orders are dispreferred, but we have seen that these orders do occur in many cases. Returning to the Neo-Emergentist model presented above and focussing on the verbal domain, the prediction is that a verbal category is postulated first before more fine-grained distinctions can emerge. These distinctions are notable when there seems to be a systematic departure from Saussurean arbitrariness between sound-meaning mappings Biberauer (2019), although phonological and semantic cues can also be noticed. For example, English modal verbs like *can* arise in similar position when finite and show related agreement patterns in having a past tense form, but lack an infinitive, lack 3rd-person singular *-s*, and can raise in questions. This suggests that modal verbs like *can* pattern as a subset of lexical verbs, and an additional category must be postulated to acquire these

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<sup>10</sup>This may be misleading in some cases. For example, different types of leaking OV phenomena can be accounted for under the Neo-Emergentist model, but here the ‘head-finality’ should instead be acquired as a feature on a subset of object heads (Biberauer, 2017)

into a system. By contrast, SFPs in Chinese varieties<sup>11</sup> have characteristics such as small size, less grammaticalisation, more semantic bleaching and freedom in positioning, lack of agreement, and even phonological similarities (Song, 2019). Without committing to any of these as an individual cause, an acquirer may well not acquire these SFPs into the system as a subset of verbs, and given these differences it is unsurprising these are less affected by the pressures above that might be responsible for FOFC than the English modals. This system which has little flexibility at the early stages and then can change in non-linear ways is a Complex Adaptive System in the study of dynamical systems (see Bosch (2022) for a linguistic introduction to these systems). In particular, the class of linkers proposed by Philip (2013) are noteworthy in that they do not provide any new semantic information, but just a structural and phonological reinforcement of an existing relation. Philip proposes that these must be linearised harmonically with an immediately dominating head. In this way, they serve as a boundary between semantically distinct categories without adding semantic information by themselves, and this may be useful for an acquirer to delimit relevant chunks in the strings. This shows how disharmony, even in a superficially FOFC-violating manner, may serve a useful purpose under certain conditions.

The ideas suggested above abstract away from the specific details in the papers presented, but can fit in with the points that these raise. While I have focused on Extended Projections, the likely motivation for organising structure into these is in maximising cognitive efficiency by having structure building proceed in natural chunks. This is equivalent in purpose to Spell-Out domains as highlighted by Duguine and phases as highlighted by Erlewine. The debate between statistical and absolute FOFC divides many of these papers, but this depends on the level of granularity with which FOFC is seen at. With a fine-grained set of relations between categories, much deeper than has been presented here, the linearisation patterns might look absolute and structured, but when looking across the surface, these trends appear statistical. Similarly, the pressures highlighted here necessarily coexist with other factors in the language system, which receive more attention in the form of OT constraints or other third-factors in human cognition. Overall, a wide range of approaches to FOFC are picking up on the same ideas, and these are being explored further with empirical and experimental data.

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<sup>11</sup>Or even certain emojis in the written modality in English and other languages (Song, 2019).

## 3 Range of FOFC data

### 3.1 Further points of interest

#### 3.1.1 Languages studied

Most of the early work on FOFC derives from comparative studies of Germanic, in which five permutations of Verb, Object, and Tense are variously observed, but the FOFC violating order is not. These facts are curious on their own, but it has been shown that FOFC extends to other languages, allowing for a plausible universal motivation. Languages like German are atypical in relation to their headedness of major categories, and this may have influenced the development of research strands. The verbal domain is primarily head final, although complementisers are always initial and finite verbs in matrix clauses raise to Verb Second (V2) position. The nominal domain has overt initial articles, and primarily prepositions (although some postpositions too), which do not extrapose rightwards so surface preverbally. This emphasises a divide between the nominal and verbal extended projections, which are more closely correlated in the majority of languages.

Some of the other major languages studied in research into FOFC include Finnish, Chinese varieties, and Basque. The original paper on FOFC investigates Finnish, which has relative freedom between the order of Tense, Verb and Object categories, depending on information structure, apart from an extremely limited acceptability of the FOFC violating pattern (Holmberg, 2000). As has already been seen, Chinese varieties, particularly standard Mandarin, are the most spoken and well-known languages which pose a surface challenge to the universality of FOFC with their SFPs. The variety of these has seen much research into their scope and positioning in hierarchies, as well as their characteristics. Basque allows for disharmonic orders and a variety of movement possibilities, and notably allows alternations between phonetic realisation of certain elements. This has made it a key language into research between PF realisations and FOFC (Etxepare and Haddican, 2017).

Data which collaborates different accounts of FOFC has been reliably found across a wide typological range of languages, suggesting there is something of interest which is universal here. We have focussed on Chinese SFPs, but similar particles exist in a wide variety of languages in several continents, and have been extensively analysed by Biberauer (2017). The conclusion, as was reached above, is that these do not form a uniform category, or even a neat category of exclusion, but do consistently lack properties typical of the FOFC-respecting auxiliaries found in languages like English. Further data investigating FOFC has taken place in Romance languages (Ledgeway, 2012; Myler, 2020), West African languages (Hein and Murpy, 2022; Aboh, 2020), Amahuaca (Clem, 2022), Lingua di Segni Italiano (Cecchetto, 2013), and Udmurt (Asztalos, 2019), among others.

### 3.1.2 Information structure

Most work focuses on neutral word orders, as this minimises external factors. Indeed, surface FOFC-violating orders where a phrase has raised to a fronted position can be seen in English, as in the example below:

- (13) I said I would go to the pub, and go to the pub I did.

Here, there is a surface VOT order, but this does not pose a threat to FOFC as it requires non-neutral prosody and pragmatic interpretation<sup>12</sup>. Even in languages which have more grammaticalised fronting, this order can also appear. This can be seen in languages like revived Cornish, which requires fronting of the verb with a constituent in its specifier as a topic (Akademi Kernewek, Korpus Kernewek, Cornish Corpus):

- (14) *Kelli ow hi byghan a wrug vy*  
 Lose-INF my dog little PTCL did-1sg-PAST I  
 ‘I lost my little dog’

Topicalisation of VO phrases does not seem to be prevented by FOFC across languages. Another possible exception, which has been less studied, is a construction in which an object is fronted, and then a verb is separately focussed. This has been observed in Finnish (Holmberg, 2017) and Chechen (Komen, shed)<sup>13</sup>.

However, it is not necessarily anything goes when there are deviations from base generated orders. A particularly interesting case comes from Itelmen (Bobaljik, 2024). The default word order is head-final, but old information can be extraposed rightwards. This is often obscured by information structural constraints leading to fairly free word order possibilities, but responses from an elicitation task are shown below (Bobaljik’s example 39).

- (15) *qaʔm (βeqaʔnʲɬ) ʈəm-aq (\*βeqaʔnʲɬ) tʰ-iʈ-čen (βeqaʔnʲɬ)*  
 NEG (bear) kill-NEG (\*bear) 1SG.-AUX-1>3SG (bear)  
 ‘I didn’t kill (the/a) bear.’

The most neutral order is OVT, but VTO is also available when O is old information<sup>14</sup>. This appears to be rightward extraposition of the old O, which

<sup>12</sup>Although for Kiparsky (2024), such examples suggest that FOFC is only a tendency rather than a universal.

<sup>13</sup>Cinque (2023) gives examples of languages which have been argued to have FOFC violating surface strings. Information structural concerns may be able to account for this, pending further research.

<sup>14</sup>Volodin (1986) notes a few examples of TOV order as well, which have limited context but appear to be old VP information, supporting the following point (Bobaljik, 2024)

moves above the T rather than the minimal non-string vacuous movement to form a VOT order. This would have consequences for the theories of rightward dependencies suggested earlier, but it also might be the case that FOFC can be sensitive to non base-generated orders<sup>15</sup>.

### 3.1.3 Other disharmonic orders

As discussed by Hawkins (2013), we must make sure that a disproportionate focus on surface FOFC forms does not miss wider generalisations about word orders and disharmony. Reverse-FOFC forms seem generally more available in cases of leaking OV languages, but there are phenomena which pattern in the opposite way, such as the high frequency of VO languages with final question particles, when compared to OV languages with initial question particles. These phenomena seem to capture asymmetries between the high and low periphery, which have been argued here to be a key component of FOFC explanations, although do require separate formal study.

FOFC focuses on the availability of 2-3-1 orders, just one of the six possible permutations of three elements. In some cases like Greenberg’s Universal 20 on the ordering of elements in a noun phrase, and Germanic verb clusters, 2-3-1 orders are available and 2-1-3 is the missing order (Biberauer, 2018; Culbertson et al., 2012). These have in common that they both begin with the middle element rather than one of the peripheral ones, which are a more logical starting point, although 2-1-3 is not available as a basic order without movement under standard Generativist assumptions. Medeiros (2021)’s system of a Universal Parser as a Universal Grammar captures both of these restricted orders in their respective domains, as well as allowing other unusual linearisation phenomena like the limited cross-serial dependencies found in languages like Dutch<sup>16</sup>. As this proposal uses stack sorting to a universal semantic hierarchy, this suggests that cognition plays a wider role in determining which word orders are available. We have seen enough evidence that FOFC merits study independent of other linearisation patterns, but it is predicted and evidenced that the third-factor explanations for it see further use within human cognition of language and more generally, and ultimate explanations for FOFC may well be able to capture other facts about linearisation.

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<sup>15</sup>Rightward extraposition has received other explanations in literature on FOFC. Zeijlstra (2023) notes that if CPs constitute the end of an EP, then non-extrapolating initial CP complements in OV languages should be able to surface. This is not widely attested, suggesting a motivation for the head-final filter and related effects separate from FOFC. Alternatively, under the LCA, rightward extraposition could emerge as a case of FOFC-motivated lower copy spellout (Sheehan, 2009)

<sup>16</sup>An interesting example that Medeiros mentions that cannot be easily accounted for is that of mesocclisis, where an affix surfaces more peripherally to its verb base than a clitic, as in European Portuguese. Mesocclisis has also been found in limited circumstances in other romance languages, and had even been borrowed into Albanian varieties (Manzini and Savoia, 2011), meaning this requires further attention.

### 3.1.4 Morphology

FOFC has also been observed, although contentiously, in morphology. There are clear cases where suffixing follows prefixing, as in [[dis[appear]]ed], which has been argued to show that FOFC does not hold within words (Kiparsky, 2024). However, that affixation proceeds in different stages is not uncontroversial in morphological theory, raising interesting questions about the temporal relations of this. FOFC has been argued to hold in morphology (Myler, 2009; Roberts, 2017a). This must be subject to different cycles, which is similar to the syntactic occurrences for FOFC mentioned above. Within a given cycle, it is predicted that prefixes must be more peripheral than all suffixes. This has been linked to the suffixing prefix (Roberts, 2017a)<sup>17</sup>.

In particular, whether processes of nominalisation must respect FOFC is a slightly contentious issue. According to view of Extended Projections that has been presented so far, there is no reason that FOFC should hold in nominalisation. This is supported by data from Tupari in which nominalisation takes a surface FOFC-violating form (Singerman, 2021). However, using the notion of mixed extended projections (CITE), Hein and Murpy (2022) show data from word order alternations across a range of languages in West Africa that FOFC does appear to hold here in nominalisation. In languages with basic VO order and suffix nominalisers like Buli (Gur, Niger-Congo), the pattern is that the VP becomes OV when nominalised.

- (16) a. *dê mǎngò-kú-lá*  
           ate mango-DEM-DEF  
           ‘ate that mango’  
       b. *mǎngò-kú dē-kā*  
           mango-DEM eat-NMLZ  
           ‘eating the mango’

### 3.1.5 Acquisition and Multilingualism

Many studies look at the synchronic status of FOFC in the systems of monolingual adult L1 speakers. This provides the most neutral source of empirical data, but the view of FOFC presented here relies strongly on cognitive factors, and places a high importance on the role of acquisition. Several papers have focused on acquisition, diachrony and multilingualism, and the empirical data and predictions that this should have in relation to different conceptualisations of FOFC.

In child L1 acquisition of languages like English, where FOFC effects do hold,

<sup>17</sup>There may be other interesting links with morphophonological processes. As one example, in non-directional vowel harmony, there is a bias against prefix triggers whereas no such bias against suffix ones (Finley and Badecker, 2009)

FOFC-violation errors by children are not completely absent, but are rare when compared to other more typical errors (Biberauer et al., 2017). This seems to support the idea of Input Generalisation. The two potential learning paths suggested for this are the None>All>Some path of Biberauer (2017, et subq.) and the None>Some>All path of DelBusso (2020). These two can appeal to other cognitive processes. None>All>Some has been proposed for categorisation of basic colour terms (Jaspers, 2012), among others, whereas None>Some>All more directly captures the monotonicity that these patterns create. In a theory with an impoverished Universal Grammar, the two are reasonably compatible, as *All* and *Some* are relative terms in relation to the emergence of categories in a full system. There are differences in what these predict should arise in intermediate stages of underspecification, before the system is properly acquired, but gathering uncontroversial empirical evidence for this may prove difficult. None>All>Some captures directly Input Generalisation and the frequency of rigidly head-final systems, but None>Some>All can better account for diachronic change. This is because a change from 'All' head-final to 'None' head-final should require successive changes of 'Some' head-finality. This matches attested diachronic changes in Germanic (Biberauer et al., 2017) and Latin (Ledgeway, 2012). Asztalos (2019) has looked at Udmurt, an uralic language, which is changing from OV to VO under the influence of Russian. It is found there that head-initial CPs are not ubiquitous by the time lower projections appear as head-initial, which is not predicted by an absolute formulation of FOFC<sup>18</sup>. As proposed by Whitman (2008), recurrent diachronic processes may influence surface linearisation to a large extent. In particular, the diachronic origins of SFPs when compared to FOFC-obeying complementisers have been studied in varieties of Chinese and Medumba (Tyrchan, 2021). FOFC predicts that there are differences as to which disharmonic elements can be borrowed across languages. There are some examples which seem to support this. For one, the Persian complementiser *ki* has been borrowed as an initial complementiser into many languages, including OV ones like Turkish, whereas final complementisers have not been borrowed on this scale. On the other hand, SFP elements are very readily borrowable, as they have been introduced into many world Englishes, or maintained in processes of creolisation (Aboh, 2020). This suggests such elements are quite distinct from grammaticalised auxiliaries. There has been some work on the synchronic status of FOFC in bilinguals. Myler (2020) looks at the combinatorics of codeswitching possibilities in bilinguals of Spanish, which is predominantly head-initial, and Quechua, which is strongly head-final. It is predicted that there should be a disproportionate preference for head-initial structures to not violate FOFC, but it is hard to isolate this from other sociolinguistic variables to obtain conclusive results. Meng and Nakamoto (2018) study bilingual acquisition of Chinese, which has head-initial VPs, and Japanese, which is strongly head-final, in children from ages 2;1 to 5;0. The most interesting finding from a FOFC perspective is that the children can produce

<sup>18</sup>This could be a statistical effect, or the category of higher heads may differ from lower ones, as has been seen in their behaviour (Biberauer, 2017).

instances of Chinese head-initial sentences with Japanese final complementisers. This is a surprising result for essentially all formulations of FOFC, and suggests more research needs to be done on how multiple languages systems with distinct categories may interact. A different surprising case, of Italian-Lingua di Segno Italiano bilinguals, will be covered in the next section.

### 3.1.6 Modality

So far, we have only seen FOFC as cognitive biases in the externalisation of language across the auditory modality. Here, the limited capacity of information transmission increases the importance of linear sequence in minimising information loss. However, it should still be predicted that the cognitive processes underlying categorisation play a role in transmission across other modalities. Sign languages have been an important source in investigating other linearisation patterns, as their use of non-manual marking and availability of multiple articulators lead to key difference. In particular, they display (not entirely uncontroversially in different theoretical approaches) rightward wh-movement, and by extension right-linearised specifiers. The only exploration of FOFC effects in sign languages so far is in Cecchetto (2013)<sup>19</sup>. Native signers of Lingua di Segni Italiano (LSI) prefer OV order, whereas L1 Italian learners are influenced by the VO order of Italian. LSI has uninflecting final particle-like elements, signalling for example negation or aspect. Although the L2 signers generally prefer VO order, they strictly avoid it when used with a final particle, avoiding the FOFC violating surface order. Not only is this slightly surprising in relation to the other findings of sign language linearisation, but as we have seen uninflecting SFPs often are not affected by FOFC<sup>20</sup>. This is an interesting place for further investigation, as it is not clear what current approaches should predict possible. Furthermore, the same cognitive processes might interact differently with semiotics in modalities like writing, and these effects could extend beyond language.

## 3.2 Evaluation of points of interest

This section has highlighted some lines of investigation into what FOFC can teach us about wider domains in language and beyond. These support the nuanced approach taken so far. The asymmetries which were relevant for the approaches of the core of FOFC, particularly relating to bottom-up and left-right in structure building and linearisation, reoccur in these other domains. As before, there are no absolute simple solutions, but these patterns are worth exploring. Information structure is very important in extended the range of forms

<sup>19</sup>Kiparsky (2024) argues that sign languages lack functional categories and this explains why they should have SOV order (although Cecchetto gives an example of SVO order in sign language here), within the approach which suggests different constraints for lexical and functional categories.

<sup>20</sup>Some of the more significant diagnostics of SFPs in spoken languages like the absence of concatenative agreement are less relevant within sign languages. This might be a good place for research into the nature of SFPs across modalities.



and meanings and the associated mappings. The widespread use of movement means that the surface orders which are avoided as base generated, appear more freely in language use, which poses problems for some formulations of FOFC. Movement of topics and focussed elements is generally high and leftwards, although the Itelmen data is an indication that languages have some flexibility here. Other elements also have linearisation preferences, as has been seen with specifiers to the left and question marking heads to the right. Something notable about these elements is that they often do not change the category of the elements they merge with, but are used in marking the edge of it from a linear perspective. Delimitation such as this plays an important role, including prosodic bootstrapping for the earliest syntactic acquisition (Biberauer, 2017), as well as the units of words, of languages in situations of codeswitching, and of modalities.

A tentative sketch of how these factors fit together is presented here. The most basic and first acquired EPs display a preference for harmony and symmetry, but further up the EPs, at a level past where lexical items are first merged, the system becomes unpredictable. There are comparisons to be made here with the *edge of chaos* phenomenon in complex adaptive systems. The system sees progress in integrating new forms and functions. This can be seen as the duality of semantics (Chomsky, 2007), where internal merge and prosodic cues contribute to semantics above the level of lexical composition, such as information structure and pragmatic cues. Categories can be finely refined like the auxiliaries in English, and distinct elements can be straightforwardly borrowed and integrated like SFPs. At the same time, language systems do not devolve into complete chaos which would impede communication. This is guided by third factors which can organise the ontologies and contiguities between categories and their properties, such as linear order. It is also guided by cognitive factors, such as a bias for specifiers and movement to the left<sup>21</sup>, whereas particle like elements phonologically mark boundaries, and can be linearised to the right. This takes place at different levels of granularity, ranging from words, to EP, to whole systems, which can be integrated as for bilinguals, or for the use of emojis. This also resembles the Goldilocks principle, in which acquirers pick up information which is *just right* for their level, neither too simple or advanced, in other words at the edge of their current system.

Equally, the data that has been covered in FOFC can appear conflicting or confusing, as this is extended further in these studies which take a wider scope. This means that many different approaches can make hypotheses about the cognitive factors at play.

This is intended as a purposely vague point of reference for future work to outline the current connections between approaches. All of the approaches that have been reviewed here use different theoretical tools to help analyse empirical data and make novel predictions.

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<sup>21</sup>Importantly, this can also occur rightwards with marked cues such as non-manual marking in sign languages, or information structural conditions as in Itelmen.

## 4 Conclusion

In 2024, we are asking relevant questions about FOFC and there are a range of different hypotheses and formulations which can model and make predictions about different pieces of data. The discussion suggests that investigating FOFC is helping us to better understand human cognition and categorisation. FOFC seems to fundamentally arise from different asymmetries in categorisation and cognition, both bottom-up and left-right, which cannot be totally unified. These will affect language and cognition more widely, and can be investigated from many perspectives. There cannot be any agreement about a precise definition of what should be covered under FOFC, but this should not be problematic as long as care is taken to delimit the objects of individual studies and contextualise these into the wider literature. Many different theories and approaches can fruitfully study these phenomena. This review has focussed on Neo-Emergentist and Dynamical Systems theory, but the findings from classical typological, Optimality Theoretic, computational, Kaynian and processing studies, among others, have all been seen here and complement each other with their findings and analyses. In almost 25 years of study, we still cannot speak about FOFC with much clarity at all, but the traditions of research on it have and continue to be at the forefront of 21st century syntactic and cognitive theory.

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